

9D Distributed Maximum Likelihood Kalman Filter

I. SYSTEM STATE AND KINEMATICS

Consider a cooperative localization system consisting of I mobile nodes indexed by $i \in \{1, \dots, I\}$. At discrete time step t , the state of node i is defined on the manifold $\mathcal{M} = \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3)$ as:

$$\mathbf{x}_t^i = \begin{bmatrix} \mathbf{p}_t^i \\ \mathbf{v}_t^i \\ \mathbf{R}_t^i \end{bmatrix} \in \mathcal{M} \quad (1)$$

where $\mathbf{p}_t^i \in \mathbb{R}^3$ denotes position, $\mathbf{v}_t^i \in \mathbb{R}^3$ denotes linear velocity in the navigation frame (n -frame), and $\mathbf{R}_t^i \in SO(3)$ is the rotation matrix transforming coordinates from the body frame (b -frame) to the n -frame.

The IMU measurements of vehicle i consist of specific force $\tilde{\mathbf{a}}_t^i \in \mathbb{R}^3$ and angular velocity $\tilde{\boldsymbol{\omega}}_t^i \in \mathbb{R}^3$ resolved in the b -frame:

$$\tilde{\mathbf{a}}_t^i = (\mathbf{R}_t^i)^\top (\mathbf{a}_t^i - \mathbf{g}) + \bar{\mathbf{a}}^i + \mathbf{e}_{a,t}^i \quad (2)$$

$$\tilde{\boldsymbol{\omega}}_t^i = \boldsymbol{\omega}_t^i + \bar{\boldsymbol{\omega}}^i + \mathbf{e}_{\omega,t}^i \quad (3)$$

where $\mathbf{a}_t^i$ is the true acceleration in the n -frame, $\mathbf{g}$ is the gravity vector, $\bar{\mathbf{a}}^i, \bar{\boldsymbol{\omega}}^i$ are known constant sensor biases, and the measurement noises are Gaussian white sequences:

$$\mathbf{e}_{a,t}^i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_a^i), \quad \mathbf{e}_{\omega,t}^i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\omega^i) \quad (4)$$

The bias-corrected IMU signals driving the kinematic propagation are defined as:

$$\mathbf{a}_{m,t}^i \triangleq \tilde{\mathbf{a}}_t^i - \bar{\mathbf{a}}^i, \quad \boldsymbol{\omega}_{m,t}^i \triangleq \tilde{\boldsymbol{\omega}}_t^i - \bar{\boldsymbol{\omega}}^i \quad (5)$$

The Lie algebra $\mathfrak{so}(3)$ is associated with 3×3 skew-symmetric matrices via $[\cdot]_\times : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$. The exponential map $\exp : \mathbb{R}^3 \rightarrow SO(3)$ is evaluated via Rodrigues' formula:

$$\exp([\phi]_\times) = \mathbf{I}_3 + \frac{\sin \theta}{\theta} [\phi]_\times + \frac{1 - \cos \theta}{\theta^2} [\phi]_\times^2 \quad (6)$$

where $\theta = \|\phi\|$.

II. NOMINAL AND ERROR STATE PROPAGATION

A. Nominal State Propagation

Given the posterior state estimate $\hat{\mathbf{x}}_{t|t}^i$ at time step t and sampling period τ , the prior nominal state $\hat{\mathbf{x}}_{t+1|t}^i$ propagates via numerical integration:

$$\hat{\mathbf{R}}_{t+1|t}^i = \hat{\mathbf{R}}_{t|t}^i \exp(\tau [\boldsymbol{\omega}_{m,t}^i]_\times) \quad (7)$$

$$\hat{\mathbf{v}}_{t+1|t}^i = \hat{\mathbf{v}}_{t|t}^i + \tau (\hat{\mathbf{R}}_{t|t}^i \mathbf{a}_{m,t}^i + \mathbf{g}) \quad (8)$$

$$\hat{\mathbf{p}}_{t+1|t}^i = \hat{\mathbf{p}}_{t|t}^i + \tau \hat{\mathbf{v}}_{t|t}^i + \frac{\tau^2}{2} (\hat{\mathbf{R}}_{t|t}^i \mathbf{a}_{m,t}^i + \mathbf{g}) \quad (9)$$

B. Error State Dynamics and Covariance Propagation

Let the true state be parameterized by the error state components $\delta \mathbf{p}_t^i$, $\delta \mathbf{v}_t^i$, and $\delta \phi_t^i$:

$$\mathbf{R}_t^i = \hat{\mathbf{R}}_{t|t}^i \exp([\delta \phi_{t|t}^i]_\times) \quad (10)$$

$$\mathbf{v}_t^i = \hat{\mathbf{v}}_{t|t}^i + \delta \mathbf{v}_{t|t}^i, \quad \mathbf{p}_t^i = \hat{\mathbf{p}}_{t|t}^i + \delta \mathbf{p}_{t|t}^i \quad (11)$$

The discrete-time linear error dynamics of node i are:

$$\boldsymbol{\theta}_{t+1|t}^i = \mathbf{A}_t^i \boldsymbol{\theta}_{t|t}^i + \mathbf{W}_t^i \mathbf{w}_t^i \quad (12)$$

where $\boldsymbol{\theta}_t^i = [(\delta \mathbf{p}_t^i)^\top, (\delta \mathbf{v}_t^i)^\top, (\delta \phi_t^i)^\top]^\top \in \mathbb{R}^9$, $\mathbf{w}_t^i = [(\mathbf{e}_{a,t}^i)^\top, (\mathbf{e}_{\omega,t}^i)^\top]^\top \in \mathbb{R}^6$.

The state transition matrix $\mathbf{A}_t^i \in \mathbb{R}^{9 \times 9}$ and noise input matrix $\mathbf{W}_t^i \in \mathbb{R}^{9 \times 6}$ are defined compactly to fit the double-column width:

$$\mathbf{A}_t^i = \begin{bmatrix} \mathbf{I}_3 & \tau \mathbf{I}_3 & -\frac{\tau^2}{2} \hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_\times \\ \mathbf{0}_3 & \mathbf{I}_3 & -\tau \hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_\times \\ \mathbf{0}_3 & \mathbf{0}_3 & \exp(-\tau [\boldsymbol{\omega}_{m,t}^i]_\times) \end{bmatrix} \quad (13)$$

$$\mathbf{W}_t^i = \begin{bmatrix} -\frac{\tau^2}{2} \hat{\mathbf{R}}_{t|t}^i & \mathbf{0}_3 \\ -\tau \hat{\mathbf{R}}_{t|t}^i & \mathbf{0}_3 \\ \mathbf{0}_3 & -\tau \mathbf{J}_r(\tau \boldsymbol{\omega}_{m,t}^i) \end{bmatrix} \quad (14)$$

where the right Jacobian of $SO(3)$ is resolved via:

$$\mathbf{J}_r(\phi) = \mathbf{I}_3 - \frac{1 - \cos \theta}{\theta^2} [\phi]_\times + \frac{\theta - \sin \theta}{\theta^3} [\phi]_\times^2 \quad (15)$$

The discrete local process noise covariance matrix $\mathbf{Q}_t^i \in \mathbb{R}^{9 \times 9}$ and the error covariance propagation are:

$$\mathbf{Q}_t^i = \mathbf{W}_t^i \begin{bmatrix} \boldsymbol{\Sigma}_a^i & \mathbf{0}_3 \\ \mathbf{0}_3 & \boldsymbol{\Sigma}_\omega^i \end{bmatrix} (\mathbf{W}_t^i)^\top \quad (16)$$

$$\boldsymbol{\Sigma}_{t+1|t}^i = \mathbf{A}_t^i \boldsymbol{\Sigma}_{t|t}^i (\mathbf{A}_t^i)^\top + \mathbf{Q}_t^i \quad (17)$$

III. JOINT PRIOR COVARIANCE PROPAGATION

Node i maintains the joint posterior error covariance $\bar{\boldsymbol{\Sigma}}_{t|t}^i \in \mathbb{R}^{9|\mathcal{U}_i| \times 9|\mathcal{U}_i|}$ corresponding to the closed subnetwork $\mathcal{U}_i = \{i\} \cup \mathcal{N}_i$. Assuming the discrete local process noises are uncorrelated across different targets, the joint neighborhood prior error covariance $\bar{\boldsymbol{\Sigma}}_{t+1|t}^i$ and joint prior information matrix $\bar{\boldsymbol{\Gamma}}_{t+1|t}^i$ are propagated directly while preserving cross-covariances as:

$$\bar{\boldsymbol{\Sigma}}_{t+1|t}^i = \bar{\mathbf{A}}_t^i \bar{\boldsymbol{\Sigma}}_{t|t}^i (\bar{\mathbf{A}}_t^i)^\top + \bar{\mathbf{Q}}_t^i \quad (18)$$

$$\bar{\boldsymbol{\Gamma}}_{t+1|t}^i = \left(\bar{\boldsymbol{\Sigma}}_{t+1|t}^i \right)^{-1} \quad (19)$$

where $\bar{\mathbf{A}}_t^i = \text{blockdiag}(\{\mathbf{A}_t^c\}_{c \in \mathcal{U}_i})$ and $\bar{\mathbf{Q}}_t^i = \text{blockdiag}(\{\mathbf{Q}_t^c\}_{c \in \mathcal{U}_i})$. Because the nominal state has zero prior error mean, the joint prior information vector of the neighborhood error state is defined as:

$$\bar{\boldsymbol{\gamma}}_{t+1|t}^i = \mathbf{0} \quad (20)$$

IV. DISTRIBUTED GAUSS-NEWTON OPTIMIZATION

A. Problem Formulation and Variable Definition

Let $\mathcal{N}_i$ denote the set of spatial neighbors of node i . The set of nodes whose state errors are estimated by node i is defined as the local index set:

$$\mathcal{U}_i = \{i\} \cup \mathcal{N}_i \quad (21)$$

At consensus iteration k , the local state error vector maintained at node i is stacked as:

$$\mathbf{s}_k^i = \text{col} \left(\mathbf{s}_k^{i,c} \right)_{c \in \mathcal{U}_i} \in \mathbb{R}^{3|\mathcal{U}_i|} \quad (22)$$

where $\mathbf{s}_k^{i,c} \in \mathbb{R}^3$ represents node i 's estimate of the 3D position error $\delta \mathbf{p}^c$ of node c at iteration k .

The local objective function $F^i(\mathbf{s}_k^i)$ is formulated using absolute anchor measurements and relative inter-node measurements:

$$F^i(\mathbf{s}_k^i) = J_{anc}^i(\mathbf{s}_k^i) + J_{inter}^i(\mathbf{s}_k^i) \quad (23)$$

where the absolute anchor ranging cost is:

$$J_{anc}^i(\mathbf{s}_k^i) = \frac{1}{2\sigma_s^2} \sum_{l=1}^K \left(l_t^l - \left\| \hat{\mathbf{p}}_{t|t-1}^i + \mathbf{s}_k^{i,i} - \mathbf{c}^l \right\| \right)^2 \quad (24)$$

and the relative inter-node ranging cost is:

$$J_{inter}^i(\mathbf{s}_k^i) = \frac{1}{2\sigma_z^2} \sum_{j \in \mathcal{N}_i} \left(z_t^{i,j} - \left\| (\hat{\mathbf{p}}_{t|t-1}^i + \mathbf{s}_k^{i,i}) - (\hat{\mathbf{p}}_{t|t-1}^j + \mathbf{s}_k^{j,j}) \right\| \right)^2 \quad (25)$$

For each physical variable $\delta \mathbf{p}^c$, the consensus sub-network consisting of all nodes that maintain an estimate of $\delta \mathbf{p}^c$ is defined as:

$$\mathcal{U}_c = \{c\} \cup \mathcal{N}_c \quad (26)$$

B. Initialization

At iteration $k = 1$, the local variables are initialized as:

$$\mathbf{s}_1^{i,c} = \mathbf{0}, \quad \forall c \in \mathcal{U}_i \quad (27)$$

$$\mathbf{g}_1^{i,c} = \nabla_c F^i(\mathbf{s}_1^i), \quad \forall c \in \mathcal{U}_i \quad (28)$$

$$\mathbf{H}_1^{i,(c,d)} = \nabla_{c,d}^2 F^i(\mathbf{s}_1^i), \quad \forall c, d \in \mathcal{U}_i \quad (29)$$

where $\nabla_c F^i$ and $\nabla_{c,d}^2 F^i$ denote the first and second partial derivatives of the objective function F^i with respect to variables c and d .

C. Localized Weight Matrices

For each variable $c \in \{1, \dots, I\}$, the localized Metropolis-Hastings weight matrix $\mathbf{W}_c = [w_c^{i,j}] \in \mathbb{R}^{|\mathcal{U}_c| \times |\mathcal{U}_c|}$ is configured as:

$$w_c^{i,j} = \begin{cases} \frac{1}{1 + \max(d_c^i, d_c^j)} & \text{if } j \in \mathcal{U}_c \cap \mathcal{N}_i \\ 1 - \sum_{l \in \mathcal{U}_c \cap \mathcal{N}_i} \frac{1}{1 + \max(d_c^i, d_c^l)} & \text{if } j = i \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

where $d_c^i = |\mathcal{U}_c \cap \mathcal{U}_i| - 1$ represents the localized communication degree of node i within subnetwork $\mathcal{U}_c$.

Similarly, for each cross-variable pair (c, d) where $c, d \in \mathcal{U}_i$, the weight matrix $\mathbf{W}_{c,d} = [w_{c,d}^{i,j}] \in \mathbb{R}^{|\mathcal{U}_{c,d}| \times |\mathcal{U}_{c,d}|}$ is constructed over the subnetwork $\mathcal{U}_{c,d} = \mathcal{U}_c \cap \mathcal{U}_d$ as:

$$w_{c,d}^{i,j} = \begin{cases} \frac{1}{1 + \max(d_{c,d}^i, d_{c,d}^j)} & \text{if } j \in \mathcal{U}_{c,d} \cap \mathcal{N}_i \\ 1 - \sum_{l \in \mathcal{U}_{c,d} \cap \mathcal{N}_i} \frac{1}{1 + \max(d_{c,d}^i, d_{c,d}^l)} & \text{if } j = i \\ 0 & \text{otherwise} \end{cases} \quad (31)$$

where $d_{c,d}^i = |\mathcal{U}_{c,d} \cap \mathcal{U}_i| - 1$ is the local communication degree of node i within subnetwork $\mathcal{U}_{c,d}$.

D. Consensus and Local Newton Iterations

For iterations $k \geq 1$, each node i updates its local error state, gradient tracking, and Hessian tracking:

1) *State Consensus Update*: For each variable $c \in \mathcal{U}_i$:

$$\mathbf{s}_{k+1}^{i,c} = \sum_{j \in \mathcal{U}_c \cap \mathcal{U}_i} w_c^{i,j} \mathbf{s}_k^{j,c} - \alpha_k^i [\mathcal{B}(\mathbf{H}_k^i)^{-1} \mathbf{g}_k^i]_c \quad (32)$$

where α_k^i is the step-size, $\mathbf{g}_k^i = \text{col}(\mathbf{g}_k^{i,c})_{c \in \mathcal{U}_i}$, and $\mathbf{H}_k^i$ is the local block diagonal-modified Hessian constructed from blocks $\mathbf{H}_k^{i,(c,d)}$ for $c, d \in \mathcal{U}_i$. $\mathcal{B}(\cdot)$ is the positive definiteness projection map.

2) *Dynamic Gradient Consensus Tracking*: For each variable $c \in \mathcal{U}_i$:

$$\mathbf{g}_{k+1}^{i,c} = \sum_{j \in \mathcal{U}_c \cap \mathcal{U}_i} w_c^{i,j} \left[\mathbf{g}_k^{j,c} + \nabla_c F^j(\mathbf{s}_{k+1}^j) - \nabla_c F^j(\mathbf{s}_k^j) \right] \quad (33)$$

3) *Dynamic Hessian Consensus Tracking*: For each pair (c, d) where $c, d \in \mathcal{U}_i$ and $\mathcal{U}_{c,d} \cap \mathcal{U}_i \neq \emptyset$:

$$\mathbf{H}_{k+1}^{i,(c,d)} = \sum_{j \in \mathcal{U}_{c,d} \cap \mathcal{U}_i} w_{c,d}^{i,j} \left[\mathbf{H}_k^{j,(c,d)} + \nabla_{c,d}^2 F^j(\mathbf{s}_{k+1}^j) - \nabla_{c,d}^2 F^j(\mathbf{s}_k^j) \right] \quad (34)$$

E. Information Exchange Packets

At each iteration k , node i transmits the message packet $\mathcal{M}_k^{i \rightarrow j}$ to out-neighbor $j \in \mathcal{N}_i$:

$$\mathcal{M}_k^{i \rightarrow j} = \left\{ \mathcal{S}_k^{i \rightarrow j}, \mathcal{G}_k^{i \rightarrow j}, \mathcal{H}_k^{i \rightarrow j} \right\} \quad (35)$$

where the packet blocks are formulated as:

$$\mathcal{S}_k^{i \rightarrow j} = \left\{ \mathbf{s}_k^{i,c} \mid c \in \mathcal{U}_i \cap \mathcal{U}_j \right\} \quad (36)$$

$$\mathcal{G}_k^{i \rightarrow j} = \left\{ \left(\mathbf{g}_k^{i,c}, \nabla_c F^i(\mathbf{s}_{k+1}^i) - \nabla_c F^i(\mathbf{s}_k^i) \right) \mid c \in \mathcal{U}_i \cap \mathcal{U}_j \right\} \quad (37)$$

$$\mathcal{H}_k^{i \rightarrow j} = \left\{ \left(\mathbf{H}_k^{i,(c,d)}, \nabla_{c,d}^2 F^i(\mathbf{s}_{k+1}^i) - \nabla_{c,d}^2 F^i(\mathbf{s}_k^i) \right) \mid c, d \in \mathcal{U}_i \cap \mathcal{U}_j \right\} \quad (38)$$

The iteration terminates when $\|\mathbf{s}_{k+1}^i - \mathbf{s}_k^i\| < \epsilon$, yielding the converged error state estimate $\mathbf{s}_{DN}^i = \mathbf{s}_{k+1}^i$.

V. POSTERIOR INFORMATION FUSION AND STATE RETRACTION

A. Likelihood Information Extraction and State Space Lifting

Upon convergence of the Distributed Gauss-Newton (D-GN) iterations, the local consensus Hessian blocks $\mathbf{H}_{DN}^{i,(c,d)}$ represent the average measurement curvature over the subnetworks. To reconstruct the absolute total Fisher information, the joint neighborhood likelihood information matrix $\bar{\mathbf{\Lambda}}_t^i \in \mathbb{R}^{3|\mathcal{U}_i| \times 3|\mathcal{U}_i|}$ and vector $\bar{\mathbf{\lambda}}_t^i \in \mathbb{R}^{3|\mathcal{U}_i|}$ are extracted by rescaling with the subnetwork sizes:

$$\bar{\mathbf{\Lambda}}_t^{i,(c,d)} = |\mathcal{U}_{c,d}| \cdot \mathbf{H}_{DN}^{i,(c,d)}, \quad \forall c, d \in \mathcal{U}_i \quad (39)$$

$$\bar{\mathbf{\Lambda}}_t^i = \left[\bar{\mathbf{\Lambda}}_t^{i,(c,d)} \right]_{c,d \in \mathcal{U}_i} \quad (40)$$

$$\bar{\mathbf{\lambda}}_t^i = \bar{\mathbf{\Lambda}}_t^i \mathbf{s}_{DN}^i \quad (41)$$

where $\mathbf{s}_{DN}^i$ is the scale-free converged local state error.

Since the prior covariance matrix is defined over the full 9D error state of each node, a tangent space selection matrix $\bar{\pi}^i \in \mathbb{R}^{3|\mathcal{U}_i| \times 9|\mathcal{U}_i|}$ is constructed to lift the 3D position likelihood into the 9D joint error space:

$$\bar{\pi}^i = \text{blockdiag}(\pi^c)_{c \in \mathcal{U}_i} \quad (42)$$

where each local block $\pi^c \in \mathbb{R}^{3 \times 9}$ isolates the position error:

$$\pi^c = [\mathbf{I}_3 \quad \mathbf{0}_3 \quad \mathbf{0}_3] \quad (43)$$

The lifted joint neighborhood likelihood matrix $\bar{\mathbf{\Lambda}}_{\text{lifted}}^i \in \mathbb{R}^{9|\mathcal{U}_i| \times 9|\mathcal{U}_i|}$ and vector $\bar{\mathbf{\lambda}}_{\text{lifted}}^i \in \mathbb{R}^{9|\mathcal{U}_i|}$ are evaluated as:

$$\bar{\mathbf{\Lambda}}_{\text{lifted}}^i = (\bar{\pi}^i)^\top \bar{\mathbf{\Lambda}}_t^i \bar{\pi}^i \quad (44)$$

$$\bar{\mathbf{\lambda}}_{\text{lifted}}^i = (\bar{\pi}^i)^\top \bar{\mathbf{\lambda}}_t^i \quad (45)$$

B. Suboptimal Posterior Information Fusion

The joint prior information matrix is evaluated by directly inverting the joint prior covariance matrix:

$$\bar{\mathbf{\Gamma}}_{t|t-1}^i = \left(\bar{\mathbf{\Sigma}}_{t|t-1}^i \right)^{-1} \quad (46)$$

The joint posterior information matrix $\bar{\mathbf{\Gamma}}_{t|t}^i \in \mathbb{R}^{9|\mathcal{U}_i| \times 9|\mathcal{U}_i|}$ and information vector $\bar{\gamma}_{t|t}^i \in \mathbb{R}^{9|\mathcal{U}_i|}$ are then fused additively:

$$\bar{\mathbf{\Gamma}}_{t|t}^i = \bar{\mathbf{\Gamma}}_{t|t-1}^i + \bar{\mathbf{\Lambda}}_{\text{lifted}}^i \quad (47)$$

$$\bar{\gamma}_{t|t}^i = \bar{\gamma}_{t|t-1}^i + \bar{\mathbf{\lambda}}_{\text{lifted}}^i = \bar{\mathbf{\lambda}}_{\text{lifted}}^i \quad (48)$$

The joint posterior covariance matrix and error state vector are computed as:

$$\bar{\mathbf{\Sigma}}_{t|t}^i = \left(\bar{\mathbf{\Gamma}}_{t|t}^i \right)^{-1} \quad (49)$$

$$\bar{\boldsymbol{\theta}}_{t|t}^i = \bar{\mathbf{\Sigma}}_{t|t}^i \bar{\gamma}_{t|t}^i \quad (50)$$

C. Ego State Extraction and Retraction

Let $\boldsymbol{\theta}_{t|t}^i \in \mathbb{R}^9$ denote the sub-vector of $\bar{\boldsymbol{\theta}}_{t|t}^i$ corresponding strictly to the ego node i :

$$\boldsymbol{\theta}_{t|t}^i = [(\delta \mathbf{p}^i)^\top, (\delta \mathbf{v}^i)^\top, (\delta \boldsymbol{\phi}^i)^\top]^\top \quad (51)$$

By parsing $\boldsymbol{\theta}_{t|t}^i$, the nominal prior state estimates are updated and mapped onto the manifold $\mathcal{M}$ via the retraction operator:

$$\hat{\mathbf{p}}_{t|t}^i = \hat{\mathbf{p}}_{t|t-1}^i + \delta \mathbf{p}^i \quad (52)$$

$$\hat{\mathbf{v}}_{t|t}^i = \hat{\mathbf{v}}_{t|t-1}^i + \delta \mathbf{v}^i \quad (53)$$

$$\hat{\mathbf{R}}_{t|t}^i = \hat{\mathbf{R}}_{t|t-1}^i \exp([\delta \boldsymbol{\phi}^i]_\times) \quad (54)$$